

The Dynamic Sporadic Server Algorithm

► By Oleg Kelner

- Dynamic Sporadic Server (DSS) – proxy framework for all aperiodic tasks
- Server *budget* C_s – the maximum execution budget
- Period T_s
- CPU utilization given by $U_s = \frac{C_s}{T_s}$
- Deadline Assignment: $d = t + T_s$ and $\rightarrow$ EDF
- Schedulability Analysis: $U_p + U_s \leq 1$

- **IDLE → READY:**

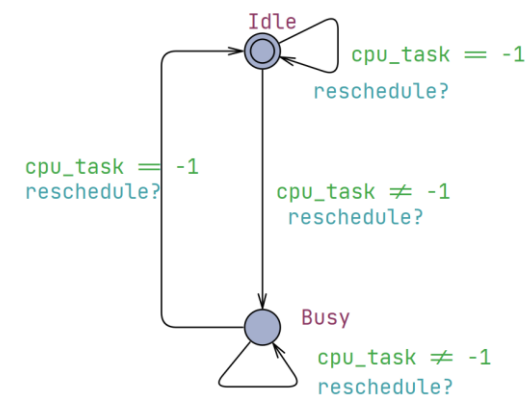
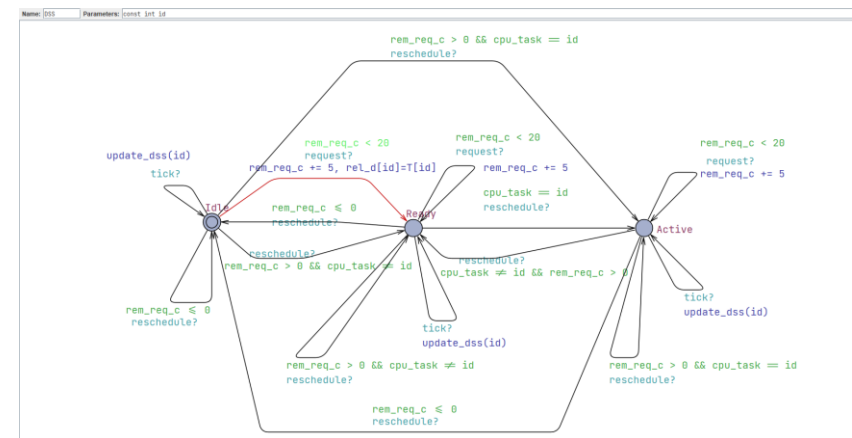
$\text{if}(C_s > 0 \ \&\& \text{pending_task}())$

$$d = t + T_s;$$

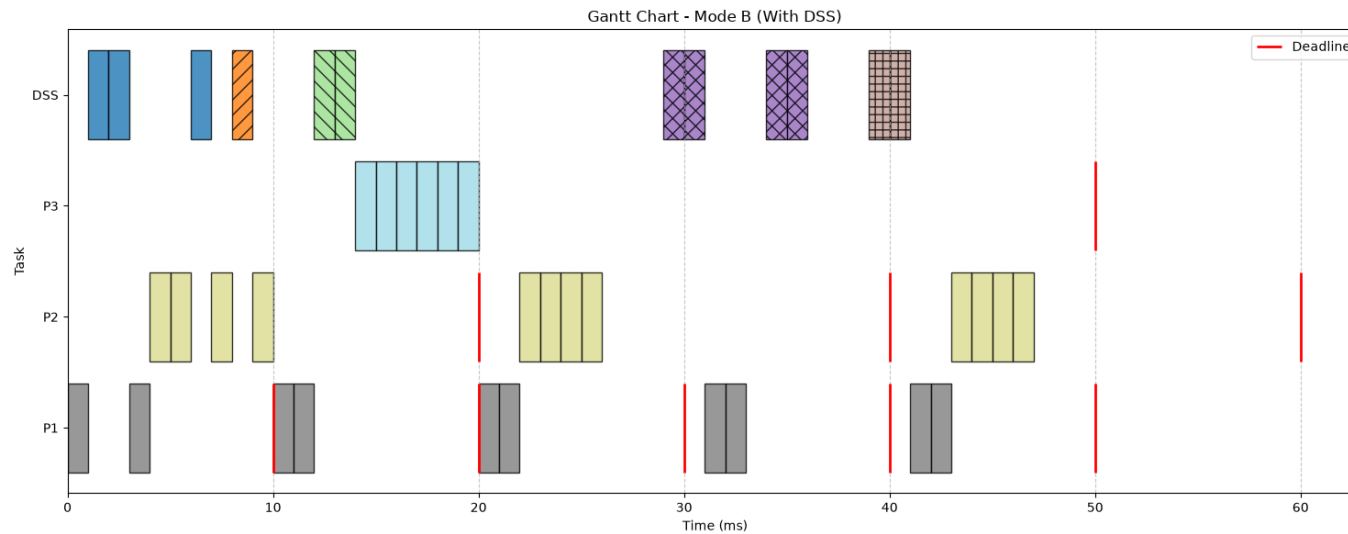
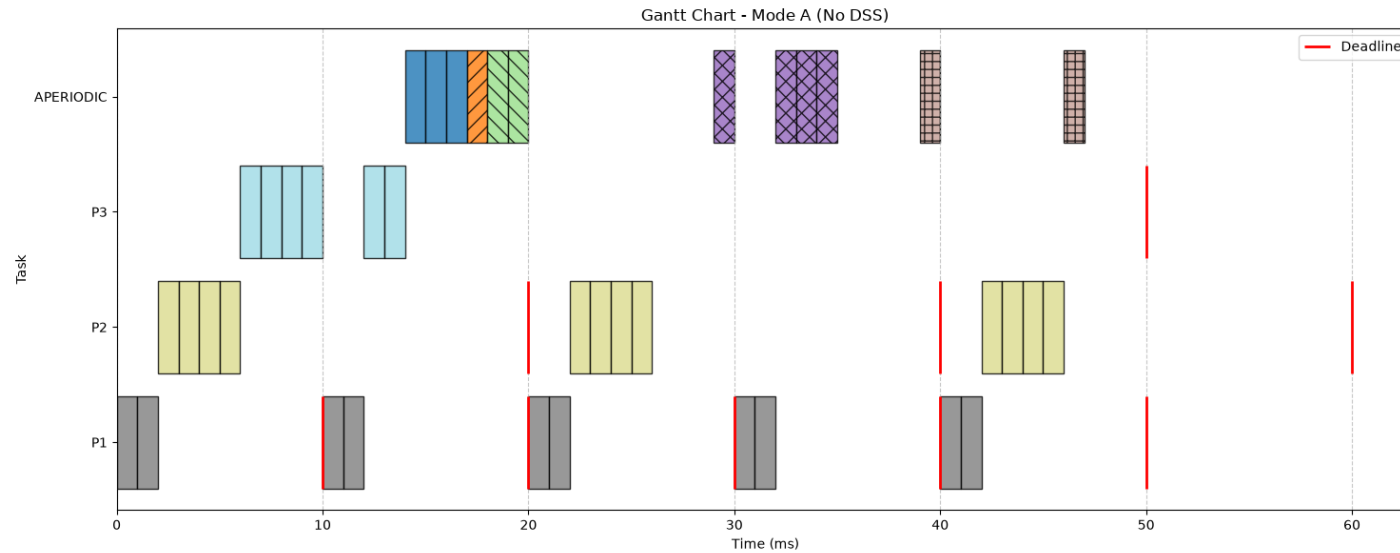
- **READY → EXE:** according to EDF
- **EXE → READY:** server is preempted by a periodic task with a shorter deadline.
- **EXE → IDLE:**

$\text{if}(C_s = 0 \ OR \ \text{task_done}())$

HOCHSCHULE
HAMM-LIPPSTADT



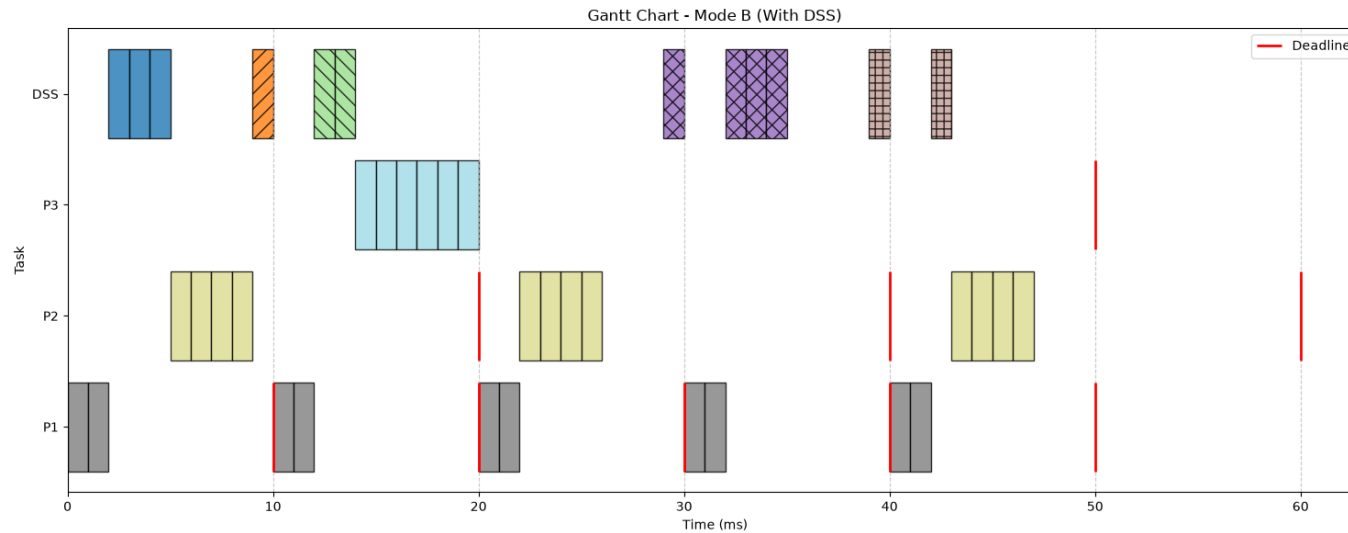
Implementation



$$C_S = 2ms$$

$$T_S = 5ms$$

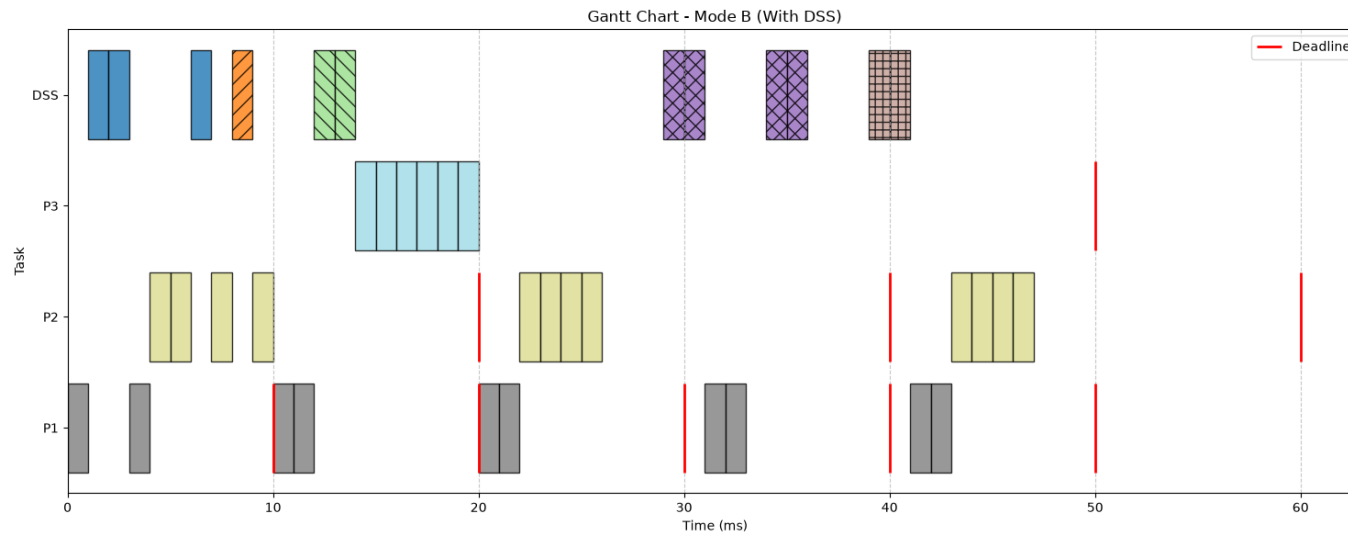
Implementation



$$C_s = 4ms$$

$$T_s = 10ms$$

$$U_s = \frac{4}{10} = \frac{2}{5}$$

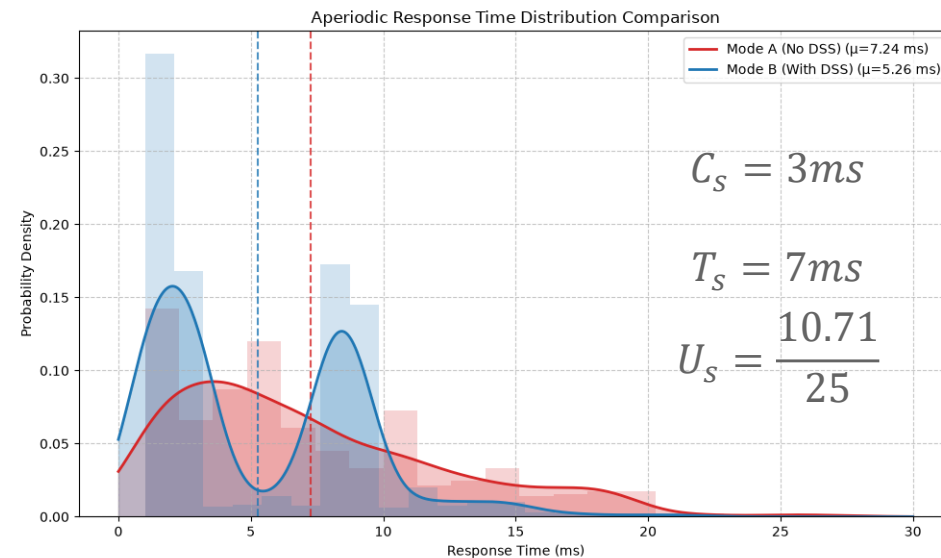
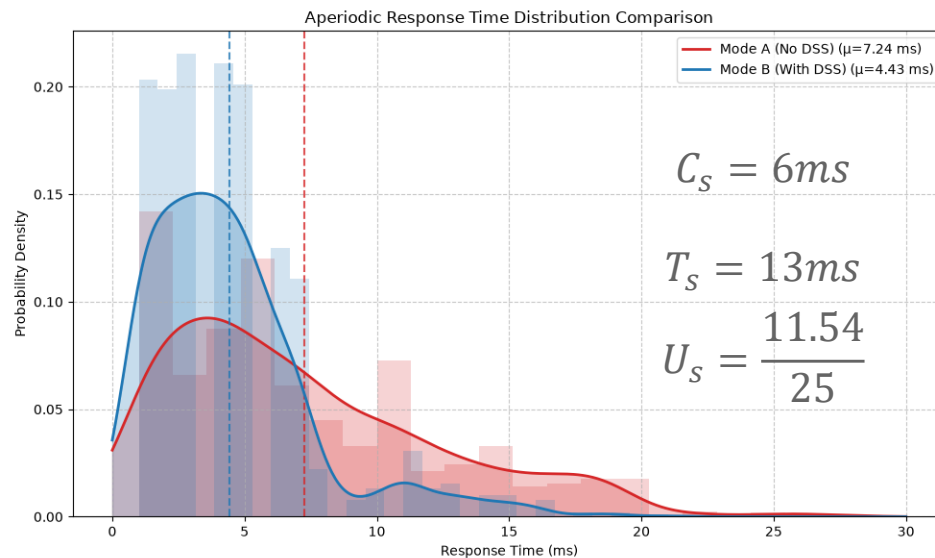
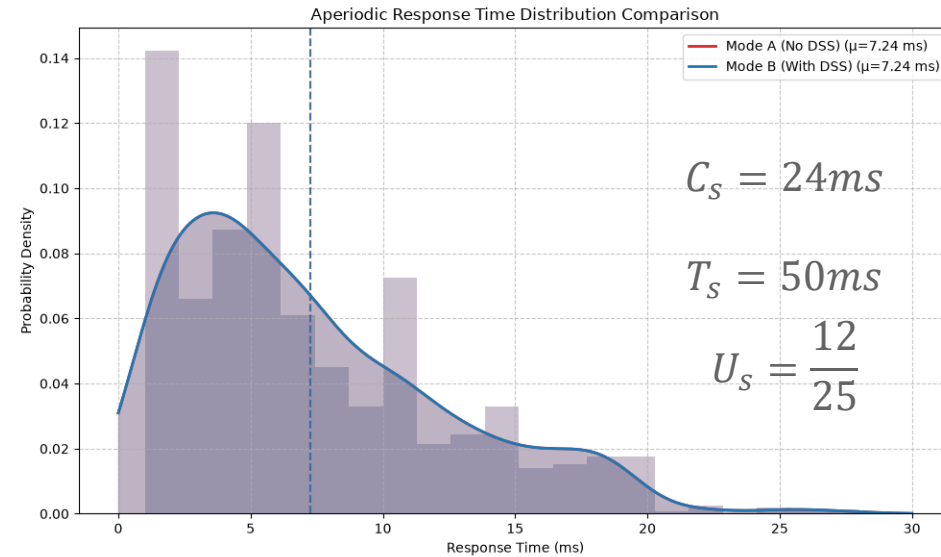
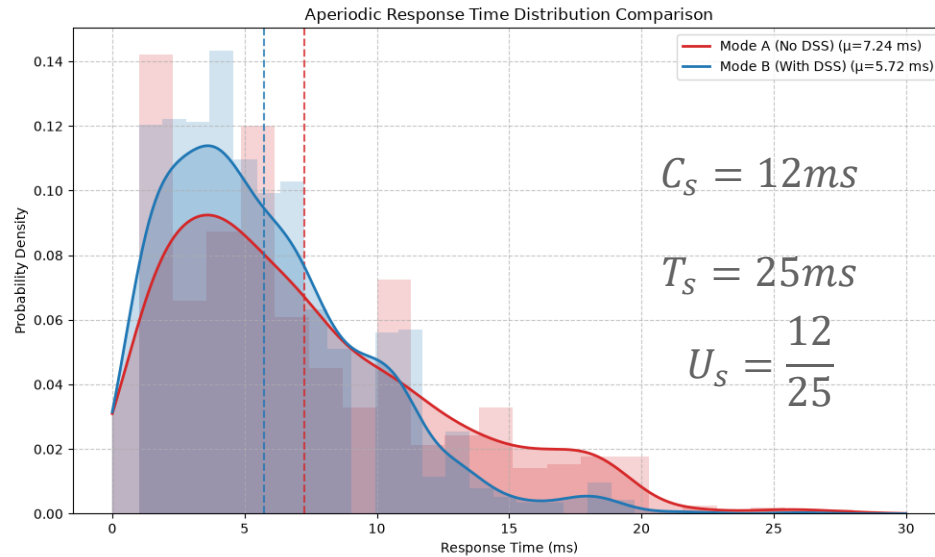


$$C_s = 2ms$$

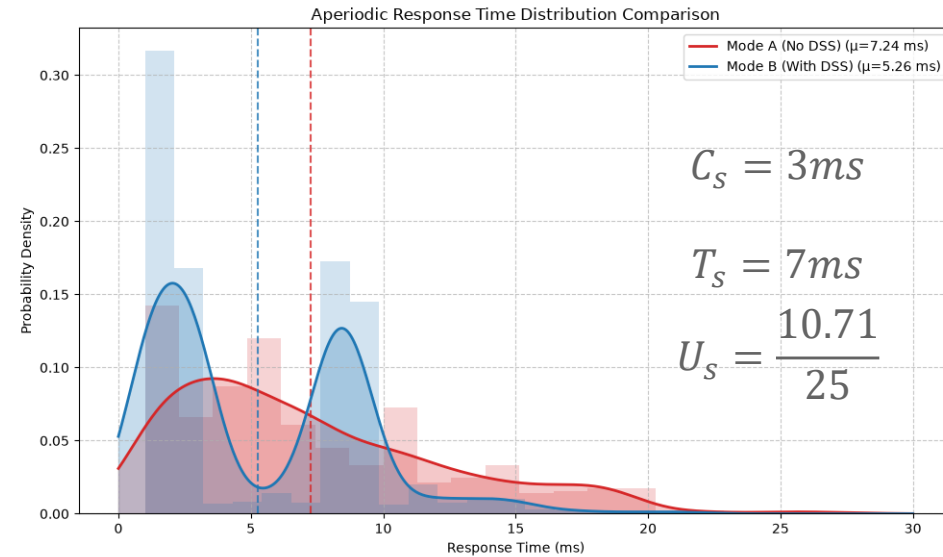
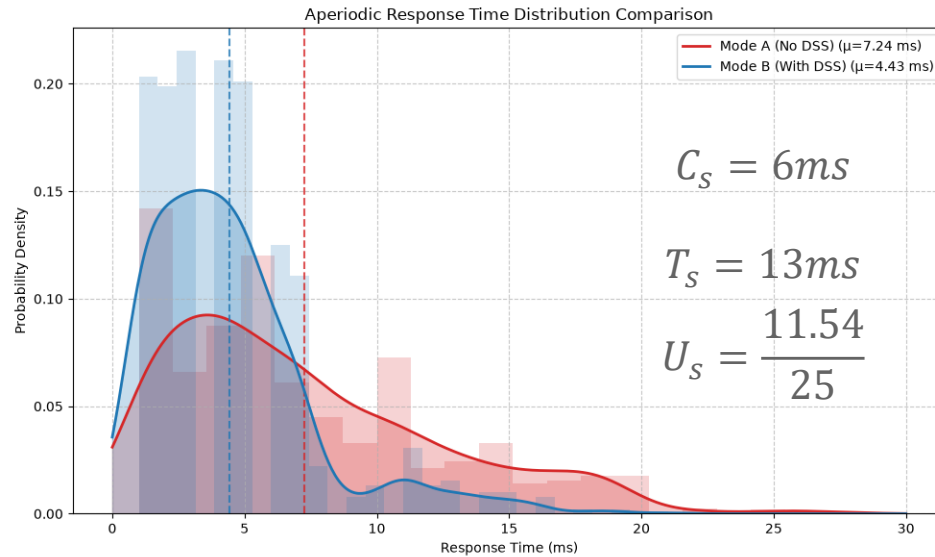
$$T_s = 5ms$$

$$U_s = \frac{2}{5}$$

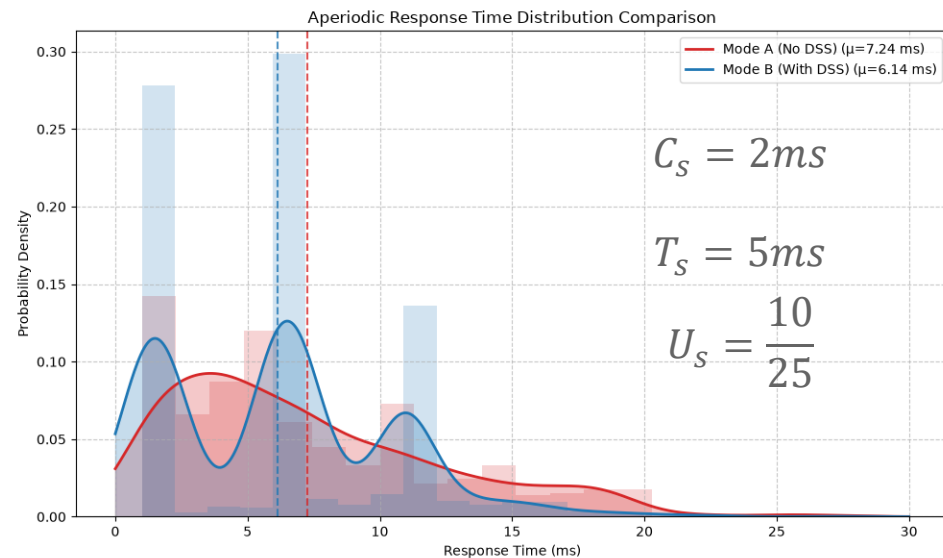
Implementation



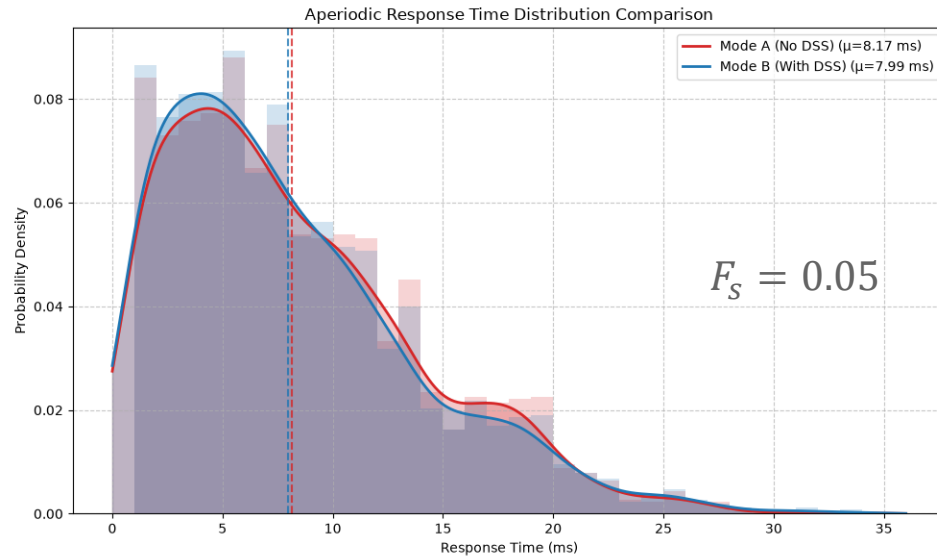
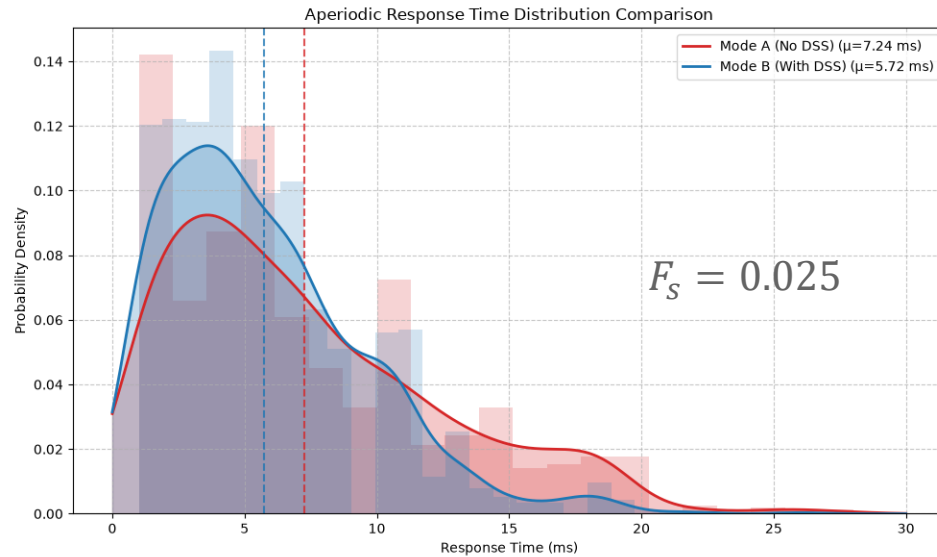
Implementation



- Reduction of T_s and C_s allow higher response time even when utilization becomes $U_s < U_{smax}$
- However, when C_s becomes smaller than task execution time, it introduces additional delay due to budget exhaustion.

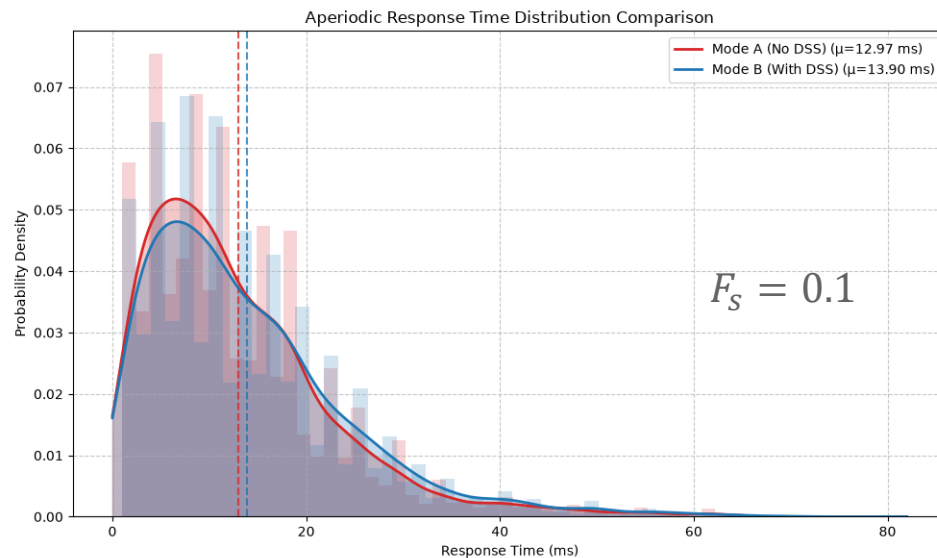


Implementation



$$C_s = 12ms$$
$$T_s = 25ms$$

- For the increased frequency of aperiodic task generation, it is clear, that at some point the performance of DSS decreases.
- Sometimes even worse than Background Scheduler



- Achieves low response times with proper configuration
- Although the calibration is highly fiddly
- Behaves great with low aperiodic task frequency

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- ChatGPT (OpenAI, GPT-5.5) was used to assist in the development of the simulation code, literature analysis, and drafting of non-technical text. All generated content was reviewed and validated by the authors.

Thank You!